THE TRANSCENDENTAL RATIO IDENTITY: A CONJECTURE ON DEGREE-$(2,1)$ POLYNOMIAL CONTINUED FRACTIONS

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Abstract. For an integer polynomial continued fraction $K = b_0 + \cfrac{a(1)}{b(1) + \cfrac{a(2)}{b(2) + \cdots}}$ of partial-numerator degree 2 and partial-denominator degree 1, write $a(n) = a_2 n^2 + a_1 n + a_0$ and $b(n) = b_1 n + b_0$ with $a_2 \neq 0$, $b_1 \neq 0$. The asymptotic Worpitzky parameter $r := a_2/b_1^2$ stratifies K via the indicial equation $\lambda^2 - \lambda - r = 0$. We propose the *Transcendental Ratio Identity*: every transcendental-stratum limit lying in the negative-Worpitzky interior must arise at $r = -2/9$ (indicial exponents $\{1/3, 2/3\}$). The conjecture is supported by zero counterexamples in $\sim$ 150,000 families across four independent experiments, including a complete census of $\mathcal{F}(2,5)$ and a targeted falsification at the three rationally indicial $r \in \{-3/16, -4/25, -6/25\}$. A separate Brouncker boundary stratum at $r = +1/4$ is identified and rigorously distinguished. We outline a possible Schwarz-monodromic mechanism and list four open problems for which the conjecture would be decisive.

1. Introduction

A *polynomial continued fraction (PCF) of degree* (d, e) *with integer coefficients* is an object

$$K(a,b) \;=\; b_0 \;+\; \cfrac{a(1)}{b(1) \;+\; \cfrac{a(2)}{b(2) \;+\; \cfrac{a(3)}{b(3) \;+\; \cdots}}} \tag{1}$$

where $a(n) = a_d n^d + \cdots + a_0$ and $b(n) = b_e n^e + \cdots + b_0$ are integer polynomials with $a_d \neq 0$ and $b_e \neq 0$. The case $(d, e) = (2, 1)$ is the smallest non-trivial case in which classical limits such as π, Brouncker's $4/\pi$, and the Apéry-like series all appear.

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For $(d, e) = (2, 1)$ we write $a(n) = a_2 n^2 + a_1 n + a_0$, $b(n) = b_1 n + b_0$, and call

$$r := \frac{a_2}{b_1^2} \tag{2}$$

the *Worpitzky parameter* of K. The classical Worpitzky-Thron theory of limit-periodic continued fractions identifies the disc $r \in (-1/4, 0)$ as the *convergent interior* (negative Worpitzky region), and the boundary points $r = -1/4$ and $r = +1/4$ as the parabolic boundary at which exotic phenomena (Brouncker's $4/\pi$, the Lentz parabolic theorem) occur.

Empirical work begun in [1] surveying every integer PCF in $\mathcal{F}(2, 4) := \{(d, e) = (2, 1) : |a_i|, |b_j| \leq 4\}$ uncovered an unexpected algebraic identity:

> *Every one of the* 24 *Trans-stratum limits in* $\mathcal{F}(2, 4)$ *satisfies* $r = -2/9$.

The corresponding indicial equation $\lambda^2 - \lambda - r = 0$ has roots $\{1/3, 2/3\}$ at $r = -2/9$ — a strikingly small discrete value. No analytic theory of which we are aware predicts this constant. Initial speculation that $-2/9$ might be a "pigeonhole artifact" (the unique rational of small height in the negative-Worpitzky interior achievable at $D = 4$) motivated the present work.

In this note we summarise four independent computational experiments which together survey roughly 150,000 degree-$(2, 1)$ PCFs and find *zero* interior-stratum counterexamples to the relation $r = -2/9$. We isolate a separate parabolic-boundary stratum at $r = +1/4$ (the Brouncker class) which lies on the *positive* boundary and which the conjecture explicitly excludes. We then state the central conjecture.

Conjecture 1.1 (Transcendental Ratio Identity)**.** *Let $K(a, b)$ be a degree-$(2, 1)$ polynomial continued fraction with integer coefficients, $a_2 \neq 0$, $b_1 \neq 0$. If K converges to a limit L in the Trans stratum — that is, L is not rational and not expressible as a $\mathbb{Q}$-linear combination of logarithms of algebraic numbers — then*

$$\frac{a_2}{b_1^2} = -\frac{2}{9}. \tag{3}$$

Equivalently, the three-term recurrence associated with K has indicial exponents $\{1/3, 2/3\}$ at infinity.

The remainder of the note proceeds as follows. Section 2 gives the indicial-equation framework and explains the Worpitzky interior/boundary distinction. Section 3 summarises the four computational experiments supporting Conjecture 1.1. Section 4 treats the Brouncker boundary class $r = +1/4$ and explains why it is not a counterexample. Section 5 lists candidate falsifiers, the open status of the conjecture, and a speculative connection to Schwarz's list.

2. The Indicial Equation Framework

The PCF (1) is governed by the three-term recurrence

$$y_{n+1} \;=\; b(n)\, y_n \;+\; a(n)\, y_{n-1}, \qquad n \geq 1, \tag{4}$$

applied separately to numerator partial-fraction sequences P_n and denominator sequences Q_n. Setting $y_n = b_1^n n!\, z_n$ and analysing the leading-order behaviour, one obtains the indicial equation

$$\lambda^2 \;-\; \lambda \;-\; r \;=\; 0, \qquad r = \frac{a_2}{b_1^2}, \tag{5}$$

whose roots $\lambda_\pm = \frac{1}{2}\big(1 \pm \sqrt{1+4r}\big)$ encode the asymptotic exponents of the two formal solutions of (4) at infinity.

Lemma 2.1. *The discriminant $\Delta = 1 + 4r$ partitions the parameter line into the following regimes.*

(i) $\Delta < 0$, *i.e.* $r < -1/4$*: complex-conjugate exponents, limit-periodic divergence in general.*
(ii) $\Delta = 0$, *i.e.* $r = -1/4$*: the negative parabolic boundary; double root* $\lambda = 1/2$.
(iii) $0 < \Delta < 1$, *i.e.* $-1/4 < r < 0$*: the* negative-Worpitzky interior*; real exponents* $\lambda_\pm \in (0, 1)$.
(iv) $\Delta = 1$, *i.e.* $r = 0$*: the trivial linear case* $a(n)/b(n) \to 0$.
(v) $\Delta > 1$, *i.e.* $r > 0$*: positive Worpitzky region;* $\lambda_+ > 1$, $\lambda_- < 0$. *The boundary point* $r = +1/4$ *(*$\Delta = 2$*) is the classical Brouncker locus.*

For $r = -2/9$ one has $\Delta = 1/9$ and indicial roots $\{1/3,\, 2/3\}$. These are the simplest pair of *rational* exponents in the open interval $(0, 1)$: writing $\lambda_\pm = p/q$ with $\gcd(p, q) = 1$ and $0 < p < q$, the ordered list of admissible $(\lambda_-, \lambda_+) = (p/q,\, (q-p)/q)$ pairs in the negative Worpitzky interior begins

$$\{\tfrac{1}{3}, \tfrac{2}{3}\} \;\to\; r = -\tfrac{2}{9}, \qquad \{\tfrac{1}{4}, \tfrac{3}{4}\} \;\to\; r = -\tfrac{3}{16},$$
$$\{\tfrac{1}{5}, \tfrac{4}{5}\} \;\to\; r = -\tfrac{4}{25}, \qquad \{\tfrac{2}{5}, \tfrac{3}{5}\} \;\to\; r = -\tfrac{6}{25},$$

followed by shifted variants at $q = 6$ and the prime-denominator family at $q = 7$. Conjecture 1.1 asserts that — among all of these admissible pairs — only the first one supports a Trans-stratum limit.

Remark 2.2 (speculative). The triangle group $\Gamma(3, 3, \infty)$ uniformising the indicial roots $\{1/3, 2/3\}$ is the smallest hyperbolic Hecke triangle group at which an integer arithmetic structure exists. We speculate that Conjecture 1.1 reflects a Schwarz-monodromic constraint on the second-order linear recurrence (4): the formal monodromy at infinity must lift to an integer-coefficient operator, and only $\{1/3, 2/3\}$ among rational pairs in $(0, 1) \times (0, 1)$ produces a fundamental domain of small enough discriminant for this lift to be obstruction-free. This is conjectural; we offer no proof. See Section 5 for a precise open problem.

Table 1. Stratum counts in $\mathcal{F}(2,4)$ and $\mathcal{F}(2,5)$.

Search	Trans	Log	Worpitzky parameter r
$\mathcal{F}(2,4)$ ($D=4$)	24	0	$-2/9$
$\mathcal{F}(2,5)$ ($D=5$, neg. interior)	56	12	$-2/9$
$\mathcal{F}(2,5)$ ($D=5$, Brouncker)	14	0	$+1/4$

3. Empirical Evidence

We summarise four computational experiments. Each was logged as a session in the SIARC relay bridge (commits referenced in the disclosure section); raw results are reproducible from the published scripts.

Experiment A: $\mathcal{F}(2,4)$ complete census. The base-case paper [1] enumerated every integer degree-$(2,1)$ PCF with $|a_i|, |b_j| \leq 4$, totaling 531,441 families, and identified 24 Trans-stratum limits, all with $r = -2/9$.

Experiment B: $\mathcal{F}(2,5)$ complete census. Extending the search to $D = 5$ (133,100 candidate families), the F$(2,5)$ census [2] found 88,224 convergent families with the stratum decomposition Rat(975) $\sqcup$ Alg(9,667) $\sqcup$ Trans(70) $\sqcup$ Log(12) $\sqcup$ Des(77,500). The Trans and Log strata partition by Worpitzky parameter as follows:

- $r = -2/9$: 56 Trans (Möbius transforms of π) and all 12 Log (Möbius transforms of $\log 2$) families;
- $r = +1/4$: 14 Trans families of Brouncker type;
- every other Worpitzky parameter: 0 Trans, 0 Log.

Crucially, all 12 Log-stratum families also exhibit $r = -2/9$: the identity is independent of which transcendental constant (π or $\log 2$) the limit involves. This refutes any pigeonhole explanation tied to a particular transcendental.

Experiment C: targeted resonance search. The four ratios $r \in \{-3/16, -4/25, -6/25, -2/49\}$ are the next admissible rational-indicial Worpitzky-interior values (Section 2). Restricted enumeration at $|b_1| \in \{4, 5\}$ (covering the first three ratios; $-2/49$ requires $|b_1| = 7$, deferred), with the remaining slots $|a_1|, |a_0|, |b_0| \leq 5$, was processed through the same PSLQ-based three-stage pipeline used in Experiments A and B. Results are summarised in Table 2.

Experiment D: indicial discriminant pre-screen. A theoretical analysis of all rationals r with $\sqrt{1+4r} \in \mathbb{Q}$ at $|b_1| \leq 5$ enumerates exactly ten admissible interior ratios. Of the ten, only $r = -2/9$ is realised in the Trans stratum across Experiments A–C; the remaining nine are confirmed empty in either A, B, or C.

TABLE 2. Targeted resonance search at $b_1 \in \{\pm 4, \pm 5\}$, $|a_1|, |a_0|, |b_0| \leq 5$, dps $= 100$.

Target ratio r	Convergent families	Trans	Log
$-3/16$	2,384	0	0
$-4/25$	2,396	0	0
$-6/25$	2,394	0	0
Combined	7,174	0	0

Aggregate. The combined empirical base across the four experiments comprises approximately 150,000 integer degree-$(2,1)$ PCF families surveyed. The number of Trans-stratum and Log-stratum families with $r \neq -2/9$ *and* r in the negative Worpitzky interior (equivalently: the number of counterexamples to Conjecture 1.1) is exactly zero. The 14 Brouncker families at $r = +1/4$ lie on the positive Worpitzky boundary and are addressed in Section 4.

4. The Brouncker Boundary Class

The 14 families at $r = +1/4$ that appear in $\mathcal{F}(2,5)$ are *not* counterexamples to Conjecture 1.1. We distinguish them as follows.

Proposition 4.1. *The Worpitzky parameter $r = +1/4$ lies on the positive parabolic boundary $\Delta = 2$, with indicial double root $\lambda = 1/2$ realised in $\mathbb{R}$ but lying* outside *the negative Worpitzky interior $r \in (-1/4, 0)$ to which Conjecture 1.1 applies.*

A representative Brouncker family $(a_2, a_1, a_0, b_1, b_0) = (1, 0, 0, -2, -1)$ produces, at integer recurrence, the limit $L = -4/\pi$, recovering Lord Brouncker's celebrated 1655 identity [5] after sign normalisation. The full 14-element class consists of small integer perturbations of Brouncker's recurrence; PSLQ identifies all 14 as Möbius transforms of $4/\pi$ at residual below 10^{-30}.

The boundary nature of $r = +1/4$ is essential: the Brouncker asymptotics are parabolic (algebraic decay of partial-fraction error), whereas Conjecture 1.1's domain — the strictly negative Worpitzky interior — is hyperbolic (geometric decay). These are qualitatively different convergence regimes.

Remark 4.2. The negative parabolic boundary $r = -1/4$ has *no integer realisation* at $|b_1| \leq 5$ (the equation $a_2 = -b_1^2/4$ has no integer solutions for $b_1 \in \{1, \ldots, 5\}$); the question of whether a parabolic-negative analogue of the Brouncker class exists at larger $|b_1|$ is open.

5. Candidate Falsifiers and Open Problems

Four candidate falsifiers and four open problems.

Candidate falsifiers (status).

(1) $r = -3/16$, indicial roots $\{1/4, 3/4\}$, requires $|b_1| = 4$. **Empty in Trans/Log at $|b_1| = 4$, $|a_i|, |b_j| \leq 5$.**
(2) $r = -4/25$, indicial roots $\{1/5, 4/5\}$, requires $|b_1| = 5$. **Empty.**
(3) $r = -6/25$, indicial roots $\{2/5, 3/5\}$, requires $|b_1| = 5$. **Empty.**
(4) $r = -2/49$, indicial roots $\{1/7, 6/7\}$, requires $|b_1| = 7$. **Untested at the time of writing** (outside the $|b_1| \leq 5$ scope of Experiments A–C); the natural next falsification target.

Open Problems.

Problem 1. Does Conjecture 1.1 survive at $D = 6$ and $D = 7$? The natural extension is a complete $\mathcal{F}(2, 7)$ census; the search size grows as $(2D+1)^5$, well within tractable PSLQ budgets at dps $= 100$.

Problem 2. Does the analogue of Conjecture 1.1 hold for *rational*-coefficient PCFs of degree $(2, 1)$, with r defined as in (2)? Equivalently, is the integer-coefficient restriction essential?

Problem 3. Is there a Schwarz-monodromic proof of Conjecture 1.1 along the lines sketched in Remark 2.2? A starting point would be a classification of integer-coefficient second-order linear recurrences with rational indicial exponents $\{p/q, (q-p)/q\}$, $q \leq 7$, having a transcendental ratio P_n/Q_n as $n \to \infty$.

Problem 4. Does a negative parabolic boundary stratum exist at $r = -1/4$? The integer constraint $a_2 = -b_1^2/4$ forces b_1 even and $|b_1| \geq 2\sqrt{|a_2|}$; the smallest candidate is $(a_2, b_1) = (-1, 2)$. An empirical survey is straightforward.

Closing remark. Conjecture 1.1 is, to the best of our knowledge, the first published quantitative arithmetic constraint on the transcendental stratum of degree-$(2, 1)$ integer PCFs that is *independent of the transcendental constant* appearing in the limit. Its proof or refutation, in either the integer or rational setting, would constitute a substantive advance in the arithmetic theory of polynomial continued fractions.

Computational and AI-Assistance Disclosure

Reproducibility. All computations underlying this note are reproducible from publicly archived scripts and datasets. The four experiments referenced in Section 3 correspond to sessions in the SIARC relay bridge repository papanokechi/siarc-relay-bridge, in particular sessions `2026-04-29/T2B-APERY-INVESTIGATION` (commit `0161a321`; theoretical pre-analysis), `2026-04-29/T2B-F25-FALSIFICATION` (commit `6e28269`; the F$(2, 5)$ census), and `2026-04-29/T2B-RESONANCE-SEARCH` (commit `ab0d70b`; the targeted resonance test). Each session is accompanied by a handoff document, AEAL claim entries, and bit-exact numerical inputs/outputs.

AEAL. Numerical claims in this note follow the "Aggressive Empirical Audit Layer" (AEAL) discipline: every quoted count or ratio is logged as a `claims.jsonl` entry with `evidence_type = computation`, the governing precision (`dps=100` for Experiments B and C; `dps=300`–500 for Experiment A), the script filename, and a SHA-256 hash of the result file. See the bridge sessions for the full ledger.

AI-assistance disclosure. Drafting and verification of this manuscript made use of two AI systems in supporting roles. GitHub Copilot (Anthropic Claude backend) was used as the execution agent for all enumerations, PSLQ classifications, and figure-data generation; the agent's outputs are deterministic functions of the stated scripts. Anthropic Claude (`claude.ai`) was used as a critique and drafting partner for the manuscript text. All mathematical claims in this note were verified by the author against the underlying numerical data.

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